

Jake Spencer W.

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Profile

IT engineer with over 6 years of experience and strong software, systems, and applied research capabilities across web, embedded systems, and biomedical projects. Seeking an engineering role while completing my undergraduate studies. Proven ability to ship production grade and business critical software, build end-to-end systems, and learn complex domains quickly. Seeking high-impact technical roles where execution and technical ownership matter. Long term goal of part-time engineering work while completing a part-time PhD in biotech/neuropsych.

Education

UTS – Bachelor in Mechatronic Engineering and Artificial Intelligence (Honours)	Expected May 2028
UTS – Bachelor in Medical Science	Expected Nov 2026
TAFE – Cert IV in Cyber Security	2020
TAFE – Cert IV in Computer Programming	2019

Experience

Crew Member, Pattysmiths – Kings Cross, Sydney Sep 2024 – Present

- Thrived in a high-paced hospitality environment, supporting daily operations during peak service periods.
- Built strong working relationships through consistent reliability, teamwork, and a positive attitude – voted by management and peers as most liked team member.

Technical Development Manager, Coreable – Coogee, Sydney Mar 2020 – Jan 2021

- Led end-to-end technical development of an incubator idea into a production-ready startup product by collaborating with UI/UX designers and business analysts.
- Designed, built, and deployed a scalable full-stack web application using MySQL, GCP, TypeScript, Node.js, GraphQL, ReactJS, and Bootstrap.
- Owned system architecture and deployment decisions, ensuring reliability at launch by performing code reviews and managing CI/CD pipelines.

Software Developer, ClockOn – Gosford, Central Coast Jul 2019 – Feb 2020

- Worked as part of a dynamic team building web and mobile application and solutions using technologies such as C# .NET, Visual Studio, JavaScript and third-party APIs.
- Gathered requirements from analyzing technical and business perspectives and producing technical documentation.
- Ran tests, and resolved technical issues, educating management on new industry technologies to produce cross platform high-quality cutting-edge products.

Fiber Splicer, Shockman – Regional NSW 2015

- Supported telecommunications infrastructure across regional sites.
- Troubleshoot and maintained network systems in live, customer-facing environments to ensure service continuity.

Placement, Academy of Interactive Entertainment – Ultimo, Sydney 2014

- Selected for a competitive game development work experience program.
- Studied 3D animation using Blender and game development in Unity with C#, building early technical and creative skills.

Projects

VS-Code Arduino IntelliSense

- Improved developer experience for over 250 developers by automating the inclusion of compile-time C++ type definitions into Arduino workspaces during code editing and minimizing latency by caching source files.

Alternipedia

- Built a multi-perspective encyclopedia platform that enables users to compare ideological interpretations of the same topic, improving online content transparency, engagement and training sets for artificial intelligence.

Medicamina

- Designed and built a precision medicine platform providing clinicians and families with patient-specific insights, leveraging Microsoft Azure database driven analytics and dynamic data visualization.
- Developed a deep learning blood cell classifier achieving 99.5% validation accuracy, integrating image preprocessing and MLP and CNN-based classification.

Low Earth Orbit object tracker

- Designed and implemented a real-time ISS tracking system using Arduino, GPS, Wi-Fi, and public APIs, integrating orbital data retrieval and 3D-printed hardware to accurately track the International Space Station from any location.

Skills

Project Management: Agile, Waterfall, Scrum, Kanban, Software Development Lifecycle (SDLC), Gantt Charts, Unified Modelling Language (UML)

Back-end: SQL (MySQL, T-SQL, MariaDB, PostgreSQL), Supabase, GraphQL, RESTful API, Vercel, Google Cloud Platform (GCP), Microsoft Azure Cloud, Node.js, C#.NET, Java, S3 Buckets, NoSQL (MongoDB)

Front-end: CSS, SCSS, React (Next.JS), Flutter, Angular, HTML, JavaScript

Testing: JUnit, Jest, Blackbox and Whitebox testing

Systems Programming: Rust, C++, UNIX kernel system calls

Machine Learning: Python, TensorFlow, Keras, Scikit, Pandas, Neural Networks, KNIME, Ensemble Methods

Networking: Routing and Switching, OSI model, Access Control Lists (ACL) & Firewalls, Wireshark, Splunk, Metasploit, nmap, TCP vs UDP, DNS configuration, Ipv4, Subnetting, OWASP

DevOps: Git Version Control, Virtual Machine Management & Docker, Object Oriented Programming, Asynchronous Programming, Debugger and Debugging, Software Troubleshooting, Technical Leadership

Engineering: Autodesk Fusion (CAD), 3D printing, DC circuit analysis, Microcontrollers, Soldering, Mathematics (Differentiation, Partial Derivatives, Definite and Indefinite Integrals, Linear Algebra, Eigenvalue and Eigenvectors) and Statistics (Mean, Mode, Median, Permutations and Combinations, Intersections), Requirements Analysis

Biology: Cell Culturing, Microscopy, Hemocytometry, Titrations, Centrifugation, Gel Electrophoresis, Spectrophotometry, Gene Recombination (CRISPR/Cas9), Flow Cytometry, Karyotyping, Induced Pluripotent Stem Cells (iPSCs), Standard Curves, Fiji

Licenses

- Responsible Service of Alcohol and Responsible Conduct of Gambling (RSA/RCG)
- Driver's License

Special Considerations

During university semesters I'm available part time, for the remaining 26 weeks of the year I can work full time.